Functional Equations — problem 11.17

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Let f be a real-valued function defined for all real numbers x such that, for some positive constant a, the equation

$$f(x+a) = \frac{1}{2} + \sqrt{f(x) - f^2(x)}$$

holds for all x.

(a) Prove that the function f is periodic, i.e., there exists a positive number b such that $f(x+b) = f(x)$ for all x.

(b) For $a = 1$, give an example of a nonconstant function with the required properties (IMO 1968).